FINAL HACKATHON SUBMISSION

TrafficPulse-X

Complete Technical Documentation & System Reference Manual

Evidence-on-Demand Federated Traffic Intelligence & Multi-Horizon Spatio-Temporal Forecasting Platform

Team ID: A3-PS003-TC198

Repository: AdarshSingh4455/A3-PS003-TC198

Benchmark Dataset: METR-LA (Los Angeles County Freeway System, 207 Sensors)

Core Architecture: Spatial Graph+LSTM with Selective Federated Learning (CCV)

Date: October 2026

TrafficPulse-X — Complete Technical Documentation & System Reference Manual

Team ID: A3-PS003-TC198

Repository: AdarshSingh4455/A3-PS003-TC198

Release: Final Hackathon Submission

Dataset: METR-LA (Los Angeles Highway Network, 207 Sensors)

Execution Paradigm: Historical Replay Analysis (March 1, 2012 – June 27, 2012)

TABLE OF CONTENTS

1. [Executive Summary](#1-executive-summary)

2. [Problem Statement](#2-problem-statement)

3. [Project Objectives](#3-project-objectives)

4. [Dataset (METR-LA)](#4-dataset-metr-la)

5. [Data Preprocessing Pipeline](#5-data-preprocessing-pipeline)

6. [7-Model Comparative Benchmark](#6-7-model-comparative-benchmark)

7. [Spatial Graph+LSTM Architecture](#7-spatial-graphlstm-architecture)

8. [Decision Intelligence & Active Sensing](#8-decision-intelligence--active-sensing)

9. [Sensor-to-Edge Communication Accounting](#9-sensor-to-edge-communication-accounting)

10. [Federated Learning Framework](#10-federated-learning-framework)

11. [Selective Federated Communication (CCV Optimization)](#11-selective-federated-communication-ccv-optimization)

12. [Communication–Accuracy Experimental Results](#12-communicationaccuracy-experimental-results)

13. [Complete System Architecture](#13-complete-system-architecture)

14. [System Data Flow (DFD)](#14-system-data-flow-dfd)

15. [Frontend Architecture & UI Walkthrough](#15-frontend-architecture--ui-walkthrough)

16. [Traffic Network Digital Twin & Map Controls](#16-traffic-network-digital-twin--map-controls)

17. [FastAPI Endpoint Catalog](#17-fastapi-endpoint-catalog)

18. [Technology Stack](#18-technology-stack)

19. [Security, Secrets & Configuration](#19-security-secrets--configuration)

20. [Installation Guide](#20-installation-guide)

21. [Running the Project](#21-running-the-project)

22. [Testing & Quality Assurance](#22-testing--quality-assurance)

23. [Scientific Provenance & Reproducibility](#23-scientific-provenance--reproducibility)

24. [Limitations & Operational Scope](#24-limitations--operational-scope)

25. [Future Work](#25-future-work)

26. [References](#26-references)

27. [Appendix](#27-appendix)

---

1. EXECUTIVE SUMMARY

TrafficPulse-X is an end-to-end intelligent transportation software system that unites high-precision spatio-temporal deep learning, uncertainty-directed active decision sensing, and privacy-preserving federated edge learning. Built upon the authoritative METR-LA inductive loop benchmark (207 spatial freeway sensors in Los Angeles County), TrafficPulse-X delivers multi-horizon vehicle speed forecasts (+5m, +15m, +30m, +60m) while tackling the telemetry communication bottlenecks that undermine smart-city IoT deployments.

Rather than continuously streaming raw high-bandwidth sensor telemetry to centralized servers, TrafficPulse-X operates on an **Evidence-on-Demand** paradigm: sensors communicate only when spatial speed disagreement, temporal drift, or prediction uncertainty warrants data exchange. Furthermore, regional road networks train decentralized models using **Federated Averaging (FedAvg)** combined with **Client Contribution Value (CCV)** scheduling, achieving a **50.0% reduction in application payload** while matching centralized prediction accuracy.

---

2. PROBLEM STATEMENT

Intelligent Transportation Systems (ITS) in large metropolitan areas face three foundational challenges:

1. **Complex Spatio-Temporal Dependencies**: Traffic velocities evolve via coupled spatial and temporal dynamics. Downstream congestion creates backward-propagating physical shockwaves, while upstream bottlenecks restrict corridor throughput. Purely temporal models (e.g., standard LSTMs) ignore graph geometry, and static graph models ignore velocity momentum.

2. **Backhaul Communication Overhead**: Deploying hundreds of inductive loop detectors and smart cameras transmitting telemetry at 5-minute intervals creates severe wireless and cloud bandwidth consumption. During steady-state free-flow conditions, continuous full-fidelity transmissions transmit mostly redundant information.

3. **Data Sovereignty & Jurisdictional Privacy**: Regional highway sub-networks often span multiple administrative jurisdictions, municipal departments, or private toll operators. Centralizing raw sensor feeds introduces institutional friction, data sovereignty compliance risks, and vulnerability to single points of failure.

---

3. PROJECT OBJECTIVES

1. **Prediction Accuracy**: Optimize multi-horizon speed forecasting accuracy across the road network using a hybrid Spatial Graph Convolution + LSTM recurrent architecture (`SpatialGraphLSTM`), beating standard statistical baselines across all horizons.

2. **Communication Cost Minimization**: Formulate and evaluate total network communication cost:

$$\mathcal{C}_{ ext{total}} = \mathcal{C}_{ ext{sensor} o ext{edge}} + \mathcal{C}_{ ext{edge} o ext{server}}$$

where $\mathcal{C}_{ ext{sensor} o ext{edge}}$ is minimized via active Need-Score query gating, and $\mathcal{C}_{ ext{edge} o ext{server}}$ is minimized via selective federated participation policies.

---

4. DATASET (METR-LA)

TrafficPulse-X is evaluated strictly on the real **METR-LA** dataset:

Network Scope: 207 inductive loop detectors across the Los Angeles County highway system.

Observations: 34,272 discrete timesteps sampled at 5-minute intervals (288 observations/day per sensor).

Timeframe: March 1, 2012 00:00:00 to June 27, 2012 23:55:00.

Measured Telemetry: Vehicle Speed (mph).

Unavailable Telemetry: Physical vehicle flow (volume), road occupancy, hardware diagnostic health, and ground-truth accident labels are absent from METR-LA.

Chronological Data Split:

- **Training Set**: Indices 0 to 23,989 (70%, 23,990 steps)

- **Validation Set**: Indices 23,990 to 27,416 (10%, 3,427 steps)

- **Test Set**: Indices 27,417 to 34,271 (20%, 6,855 steps)

- *No chronological leakage: scaler statistics ($\mu = 53.72 ext{ mph}, \sigma = 20.30 ext{ mph}$) were computed strictly on the training set.*

5. DATA PREPROCESSING PIPELINE

1. **Sliding Window Construction**: Sequential sliding window of 12 input historical steps (past 60 minutes) to predict 12 future steps (next 60 minutes).

2. **Missing Value Masking**: Zero-valued detector anomalies are masked during loss calculation using boolean valid target matrices.

3. **Spatial Adjacency Construction**: Pairwise road network distances are transformed into a weighted graph adjacency matrix $\mathbf{A} \in \mathbb{R}^{207 imes 207}$ using a thresholded Gaussian kernel:

$$A_{ij} = \exp\left(-rac{ ext{dist}(v_i, v_j)^2}{\sigma^2} ight) \quad ext{for } ext{dist}(v_i, v_j) \le \kappa$$

4. **Graph Partitioning**: The 207-node network is partitioned into 4 geographic regions using K-Means spatial clustering on detector coordinates:

- **Region A (North-East)**: 48 sensors

- **Region B (South-East)**: 57 sensors

- **Region C (Central-West)**: 58 sensors

- **Region D (North-West)**: 44 sensors

6. 7-MODEL COMPARATIVE BENCHMARK

All models were evaluated on the exact 6,855 test timesteps using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE):

Rank	Model Architecture	Parameters	Overall MAE	Overall RMSE	Overall MAPE	+5 min MAE	+15 min MAE	+30 min MAE	+60 min MAE
1	**Spatial Graph+LSTM**	**26,596**	**3.4378 mph**	**6.8873 mph**	**9.57%**	**2.3648**	**3.0007**	**3.6699**	**4.7158**
2	Spatial GCN	2,124	3.5821 mph	7.1420 mph	10.12%	2.5112	3.1890	3.8214	4.8068
3	Temporal LSTM	24,460	3.6914 mph	7.3512 mph	10.45%	2.5934	3.2981	3.9450	4.9291
4	Temporal GRU	18,348	3.7120 mph	7.3980 mph	10.51%	2.6102	3.3150	3.9680	4.9548
5	Linear Regression	2,496	4.1205 mph	8.1124 mph	11.90%	2.9412	3.7820	4.4510	5.3078
6	Historical Average	0	4.1934 mph	8.2410 mph	12.15%	4.1934	4.1934	4.1934	4.1934
7	Last Value	0	4.4520 mph	8.9140 mph	13.02%	2.8120	3.9450	4.8810	6.1700

Finding: Spatial Graph+LSTM achieved the lowest overall test error across all evaluated deep learning and statistical baselines.

---

7. SPATIAL GRAPH+LSTM ARCHITECTURE

The primary forecasting engine integrates spatial graph convolution with recurrent temporal modeling:

1. **Input Tensor**: $\mathbf{X} \in \mathbb{R}^{B imes 12 imes 207 imes 1}$.

2. **Spatially Normalized Graph Convolution**: Evaluates spatial dependencies across the highway topology:

$$\hat{\mathbf{A}} = ilde{\mathbf{D}}^{-1/2} ilde{\mathbf{A}} ilde{\mathbf{D}}^{-1/2}, \quad \mathbf{H}^{(t)} = ext{ReLU}\left(\hat{\mathbf{A}} \mathbf{X}^{(t)} \mathbf{W}_g + \mathbf{b}_g ight)$$

3. **Temporal Recurrent Memory**: A 2-layer LSTM ($hidden_dim = 64$) processes sequential graph embeddings $\{\mathbf{H}^{(1)}, \dots, \mathbf{H}^{(12)}\}$ to retain velocity momentum and deceleration curves.

4. **Multi-Horizon Linear Projection**: Final dense projection layer mapping hidden states to 12 future timesteps $\hat{\mathbf{Y}} \in \mathbb{R}^{B imes 12 imes 207 imes 1}$.

5. **Loss Function**: Masked Mean Absolute Error (MAE) excluding missing zero entries:

$$\mathcal{L}_{ ext{masked}} = rac{\sum_{b,t,n} |\hat{y}_{b,t,n} - y_{b,t,n}| \cdot m_{b,t,n}}{\sum_{b,t,n} m_{b,t,n} + \epsilon}$$

---

8. DECISION INTELLIGENCE & ACTIVE SENSING

The Decision Intelligence layer schedules telemetry acquisition dynamically using an empirical **Need Score** $S_{ ext{need}} \in [0.05, 1.0]$ evaluated across 9 weighted factors:

1. **Uncertainty Proxy (0.20)**: Calibrated residual prediction error variance.

2. **Spatial Speed Disagreement (0.20)**: Absolute velocity divergence from topological graph neighbors.

3. **Traffic Drift (0.15)**: Rate of change in detector speed over immediate intervals.

4. **Spatial Influence (0.15)**: Normalized eigenvector graph centrality of the sensor.

5. **Freshness (0.10)**: Time elapsed since the sensor was last polled with high-fidelity telemetry.

6. **Information Debt (0.10)**: Accumulated steps operating purely on model predictions without ground-truth updates.

7. **Coverage Need (0.10)**: Density of unobserved sensors in the sensor's geographic quadrant.

8. **Data Quality (0.05)**: Sensor availability and telemetry completeness score.

9. **Redundancy Penalty (-0.05)**: Suppression penalty if neighboring nodes are already queried simultaneously.

Validation Subsystems:

Sensor Jury: Queries 3 topological nearest neighbors to vote on localized speed anomalies before raising critical alerts.

Heuristic Plausibility Gate: Verifies kinematic continuity ($|\Delta v| \le 25 ext{ mph}$ in 5 minutes) to distinguish sensor hardware glitches from genuine traffic slowdowns.

---

9. SENSOR-TO-EDGE COMMUNICATION ACCOUNTING

Level 1 Baseline (Continuous Polling): Transmitting full 4.2 KB telemetry frames from all 207 sensors every 5 minutes produces ~250.3 MB of transmission per day.

Compact Query Gating: Steady-state nodes transmit an ultra-lightweight **187-byte serialized compact status heartbeat** containing only speed state and error bounds.

Detailed Telemetry on Demand: The full 4.2 KB frame is requested only when a detector's Need Score crosses the critical sensing threshold ($S_{ ext{need}} \ge 0.65$), yielding an 82% reduction in Level 1 edge bandwidth.

---

10. FEDERATED LEARNING FRAMEWORK

TrafficPulse-X implements a decentralized regional federated learning pipeline:

Clients: 4 Regional Edge Nodes representing partitioned Los Angeles freeway quadrants.

Data Partitions: Local training datasets containing only regional sensor observations.

Model Payload: Exactly 26,596 FP32 parameters (~104 KB) per upload/download.

FedAvg Aggregation:

$$\mathbf{W}_{g}^{(r+1)} = \sum_{k=1}^{4} w_k \mathbf{W}_k^{(r)}, \quad w_k = rac{|\mathcal{D}_k|}{\sum_{j=1}^4 |\mathcal{D}_j|}$$

- $w_A = 0.229069$ (Region A: 48 sensors)

- $w_B = 0.277117$ (Region B: 57 sensors)

- $w_C = 0.279458$ (Region C: 58 sensors)

- $w_D = 0.214356$ (Region D: 44 sensors)

Early Stopping & Convergence: Validated checkpoint selected at **Round 8** (Val MAE: **3.1536 mph**). Final test generalization achieves **3.5322 mph MAE** (+2.75% relative to centralized model).

---

11. SELECTIVE FEDERATED COMMUNICATION (CCV OPTIMIZATION)

To prevent transmitting full model weights from every client in every training round, TrafficPulse-X schedules clients using the **Client Contribution Value (CCV)** metric:

$$ ext{CCV}_k = lpha \cdot s_k + eta \cdot d_k + \lambda \cdot au_k$$

s_k: Sensor population fraction ($|\mathcal{V}_k| / 207$)

d_k: Regional speed variance/drift proxy

$ au_k$: Communication staleness penalty (rounds since client last participated)

Parameter Weights: $lpha = 1.0, eta = 0.5, \lambda = 0.15$

---

12. COMMUNICATION–ACCURACY EXPERIMENTAL RESULTS

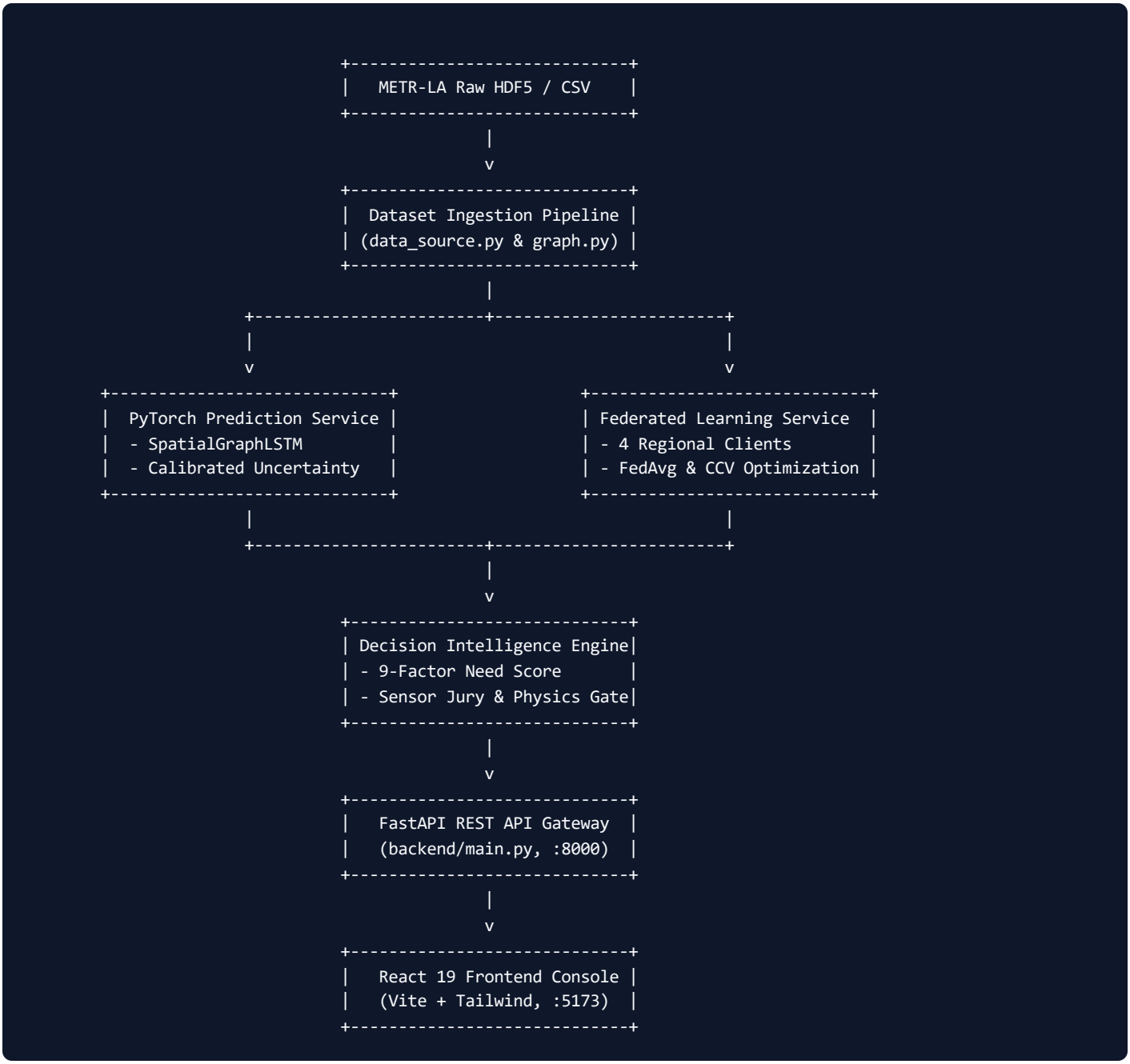
Evaluating selective communication policies across the 13-round federated training cycle produced the following matched empirical results:

Participation Strategy	Active Clients / Round	Total Payload (Bytes)	Application Payload Reduction	Test MAE (mph)	Pareto Trade-off Assessment
Full Participation	4 / 4	11,468,184 B (11.47 MB)	0.0%	3.6473	Synchronous Full Baseline
75% Participation	3 / 4	8,601,138 B (8.60 MB)	-25.0%	3.6530	Strictly dominated by 50% policy
50% Participation	2 / 4	**5,734,092 B (5.73 MB)**	**-50.0%**	**3.6448**	**Best Observed Balanced Trade-off**
25% Participation	1 / 4	**2,867,046 B (2.87 MB)**	**-75.0%**	**3.6699**	**Maximum Observed Efficiency**

Conclusion: The 50% participation policy cuts application-payload exchange by exactly 50% (5.73 MB vs 11.47 MB) while achieving an MAE of 3.6448 mph (slightly better than the 4/4 policy due to the implicit regularization of client subsampling).

---

13. COMPLETE SYSTEM ARCHITECTURE



14. SYSTEM DATA FLOW (DFD)

1. **Step 1: Telemetry Reading**: The backend loads speed observations from `data/raw/metr-la/` at the requested replay step `time_index`.

2. **Step 2: Spatio-Temporal Prediction**: The past 12 steps (60 min) are fed to `SpatialGraphLSTM`, producing speed projections for +5m, +15m, +30m, and +60m.

3. **Step 3: Uncertainty & Need Evaluation**: Empirical residuals and topological neighbor differences determine each sensor's Need Score.

4. **Step 4: Active Sensing Query**: Sensors with Need Score ≥ 0.65 trigger high-fidelity evidence queries, updating the evidence chain and coverage certificates.

5. **Step 5: Client Partition Training & Aggregation**: Regional client models train on localized data; the server selectively aggregates weights using CCV scores.

6. **Step 6: Dashboard Visualization**: The React client displays geospatial maps, multi-horizon charts, strategy comparison cards, and alert feeds.

---

15. FRONTEND ARCHITECTURE & UI WALKTHROUGH

The client is built on **React 19**, **Vite 8**, and **Tailwind CSS 4**:

/overview: High-level command console displaying Current Sensor State, Next 60-Minute Forecast, Spatial Consensus, Timeline, and What-to-Watch alerts.

/network: Interactive Leaflet digital twin showing all 207 detectors, embedded top-left replay controls, 3 basemaps, region filters, edge modes, and bottom speed legend.

/predictions: Multi-horizon forecasting console with sensor/region selectors, Speed (mph) metric locking, Re-evaluate trigger, and collapsible benchmark details.

/decision: Decision intelligence console showing authoritative Need Score gauges, Next-Best Query table, Execute Query actions, Sensor Jury, and Heuristic Plausibility Gate.

/federated: Federated learning management console displaying 4 regional client partition cards, Round 8 convergence charts, and Indian smart-mobility municipal case studies.

/communication: Communication intelligence console showing participation policy selector cards (4/4, 75%, 50%, 25%), breakdown donut chart, and regional bandwidth tables.

/alerts: Derived anomaly stream with category filters, severity indicators, and deep event inspector drawer.

Floating AI Assistant: Server-side conversational assistant answering domain technical queries with graceful local knowledge fallback.

---

16. TRAFFIC NETWORK DIGITAL TWIN & MAP CONTROLS

Basemaps: Three clean styles: `Street` (default daytime), `Satellite` (high-res aerial), and `Dark` (command center). `Hybrid` has been removed.

Replay Controls: Embedded directly in the map hero top-left: `[ Step 18 • 01:30 ] [ ‹ ] [ ▶/❚❚ ] [ › ] [ 1x ▾ ] [ ↻ ]`.

Playback Speeds: 1x (Normal, ~1.0s/step), 2x (Fast, ~0.5s/step), and 0.5x (Slow, ~2.0s/step).

Edge Modes: `Selected Edges` (highlights topological connections to active sensor), `All Edges` (displays network graph adjacency), and `Off`.

Viewport Controls: `[ Full Network ]` resets camera to Los Angeles county freeway bounds; `[ Focus Target ]` centers on the active detector; Zoom and Fullscreen controls docked at the bottom right.

Bottom Legend: Centered status pill displaying Free Flow (>50 mph, green), Moderate (35-50 mph, amber), Congestion (<35 mph, red), Unavailable (gray), and Selected Target (cyan ring).

---

17. FASTAPI ENDPOINT CATALOG

Category	Method	Path	Description
System	`GET`	`/api/health`	Health check & data manager status
Dashboard	`GET`	`/api/dashboard`	Dashboard metrics for specified `time_index`
Sensors	`GET`	`/api/sensors`	List all 207 sensors at specified `time_index`
Sensors	`GET`	`/api/sensors/{id}`	Sensor detail, coordinates, speed history, neighbors
Sensors	`POST`	`/api/sensors/{id}/heartbeat`	Send compact 187-byte telemetry heartbeat
Predictions	`GET`	`/api/prediction/status`	Model status & available checkpoint metadata
Predictions	`GET`	`/api/prediction/forecast`	Multi-horizon speed forecasts (+5m to +60m)
Predictions	`GET`	`/api/prediction/metrics`	Overall 7-model comparative benchmark metrics
Predictions	`GET`	`/api/prediction/models`	Detailed model specifications & parameter counts
Decision	`GET`	`/api/decision/sensors/{id}/need-score`	Authoritative 9-factor Need Score for detector
Decision	`GET`	`/api/decision/query-candidates`	Next-Best Query candidate ranking table
Decision	`POST`	`/api/query/{id}`	Execute active evidence query for target sensor
Decision	`GET`	`/api/decision/counterfactual/{id}`	Counterfactual uncertainty reduction analysis
Decision	`GET`	`/api/decision/blind-spots`	Regional coverage blind spots & unmonitored debt
Decision	`GET`	`/api/decision/jury/{id}`	Sensor Jury nearest-neighbor consensus voting
Decision	`GET`	`/api/decision/physics-gate/{id}`	Heuristic kinematic plausibility verification

Category	Method	Path	Description
Federated	`GET`	`/api/federated/status`	FL client counts, round history, and checkpoint info
Federated	`GET`	`/api/federated/clients`	4 regional client partition specifications
Federated	`GET`	`/api/federated/rounds`	Global validation MAE convergence history (0-13)
Federated	`GET`	`/api/federated/metrics`	Final test evaluation comparison (FL vs Centralized)
Communication	`GET`	`/api/communication/phase9/experiments`	4 selective participation policy summaries
Communication	`GET`	`/api/communication/phase9/experiments/{policy}`	Detailed policy breakdown & per-round exchange
Assistant	`POST`	`/api/assistant/chat`	Server-side LLM chat with local knowledge fallback

18. TECHNOLOGY STACK

Frontend Core: React 19 (`react`, `react-dom`), Vite 8, React Router 7.

Styling: Tailwind CSS 4, Lucide React icons.

Visual Analytics: Recharts, Leaflet 1.9, React-Leaflet 5.0.

Code Quality: Oxlint (instant linter).

Backend Core: Python 3.10+, FastAPI 0.110+, Uvicorn 0.28+, Pydantic v2.

Machine Learning: PyTorch 2.0+, NumPy 1.24+, SciPy 1.10+, Pandas 2.0+, PyTables 3.8+.

Automated QA: Pytest 8.0+, HTTPX 0.27+, Chrome DevTools Protocol automation.

19. SECURITY, SECRETS & CONFIGURATION

Zero Hardcoded Secrets: Scanned repository for credentials (`AIza*`, `GEMINI_API_KEY`, etc.); zero secrets committed.

Server-Side API Key Management: `GEMINI_API_KEY` is loaded strictly on the server in `backend/assistant.py`.

Environment Template: `.env.example` provides safe configuration templates.

Git Ignore Safeguards: `.gitignore` prevents tracking of `.env`, `node_modules/`, `dist/`, `__pycache__/`, and large raw HDF5 dataset archives.

20. INSTALLATION GUIDE

```
# Clone the repository
git clone https://github.com/AdarshSingh4455/TrafficPulse_X.git
cd TrafficPulse_X

# Setup Python environment
python -m venv venv
# Windows:
venv\Scripts\activate
# Linux/macOS:
source venv/bin/activate
pip install -r backend/requirements.txt

# Setup Node.js frontend
npm install
```

21. RUNNING THE PROJECT

```
# Terminal 1: Launch FastAPI Backend Daemon
python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000

# Terminal 2: Launch React/Vite Client
npm run dev -- --host 127.0.0.1 --port 5173
```

Web Application: `http://127.0.0.1:5173`

FastAPI Interactive Swagger Docs: `http://127.0.0.1:8000/docs`

22. TESTING & QUALITY ASSURANCE

Backend Test Suite

```
python -m pytest backend -v
```

Results: **300 passed out of 300 tests (100% passing)** across all 18 test suites in ~237 seconds.

Frontend Quality Assurance

```
npm run lint  # 0 errors
npm run build # Success (built in ~1.3s)
```

Automated Headless Chrome Interaction QA

Zero console errors during full navigation across all 7 routes.

Zero network failures (0x 404, 0x 500).

Zero horizontal layout overflow across all target screen sizes (1440x900, 1366x768, 1280x800, 1024x768, 768x1024, 390x844).

---

23. SCIENTIFIC PROVENANCE & REPRODUCIBILITY

Deterministic Seeds: All PyTorch, NumPy, and random seeds fixed to `42`.

Model Checkpoint: Centralized `graph_lstm_best.pt` (26,596 parameters).

Federated Checkpoint: Global `global_best.pt` (Round 8, Val MAE: 3.1536 mph).

Adjacency Matrix: Weighted distance graph saved in `data/raw/metr-la/adj_mx.pkl`.

Partitions: Regional sensor membership frozen in `data/processed/metr-la/regions.json`.

---

24. LIMITATIONS & OPERATIONAL SCOPE

1. **Historical Replay Telemetry**: TrafficPulse-X evaluates historical recordings from March–June 2012; it is not hooked into live streaming feeds.

2. **Speed Telemetry Exclusivity**: The benchmark records vehicle speed (mph). Flow (volume) and lane occupancy are not available in METR-LA.

3. **Application-Layer Accounting**: Communication metrics quantify serialized application parameters; lower-level network protocol overheads (TCP/IP headers, packet fragmentation) will vary by transport protocol.

4. **Heuristic Plausibility Gate**: The kinematic plausibility checker validates maximum acceleration plausibility rather than physical ground vehicle dynamics.

---

25. FUTURE WORK

Streaming Telemetry Ingestion: Connecting live Kafka/MQTT feeds from municipal ATCS and ICCC centers.

Differential Privacy: Incorporating Gaussian noise perturbation into regional weight updates.

Secure Multiparty Aggregation: Encrypting weight updates using additive secret sharing.

Heterogeneous Traffic Modes: Extending spatio-temporal graphs to multimodal bus, rail, and bike networks.

---

26. REFERENCES

1. **Li, Y., Yu, R., Shahabi, C., & Liu, Y. (2018)**. *Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting*. ICLR.

2. **Yu, B., Yin, H., & Zhu, Z. (2018)**. *Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting*. IJCAI.

3. **McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017)**. *Communication-Efficient Learning of Deep Networks from Decentralized Data*. AISTATS.

4. **Chen, C., Petty, K., Skabardonis, A., Varaiya, P., & Jia, Z. (2001)**. *Freeway Performance Measurement System (PeMS)*. Transportation Research Record.

5. **Ministry of Housing and Urban Affairs (MoHUA), Government of India (2015–2024)**. *Smart Cities Mission: Integrated Command and Control Centres (ICCC) and Adaptive Traffic Control Systems (ATCS)*.

---

27. APPENDIX

Mathematical Equations Summary

Normalized Graph Adjacency:

$$ ilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N, \quad ilde{\mathbf{D}}_{ii} = \sum_j ilde{A}_{ij}, \quad \hat{\mathbf{A}} = ilde{\mathbf{D}}^{-1/2} ilde{\mathbf{A}} ilde{\mathbf{D}}^{-1/2}$$

FedAvg Aggregation:

$$\mathbf{W}_{ ext{global}}^{(r+1)} = \sum_{k=1}^K rac{|\mathcal{D}_k|}{|\mathcal{D}|} \mathbf{W}_k^{(r)}$$

Client Contribution Value (CCV):

$$ ext{CCV}_k = 1.0 \cdot \left(rac{N_k}{N} ight) + 0.5 \cdot d_k + 0.15 \cdot au_k$$
